

📘 AI/ML Internship – Day 28 Notes & Tasks

🟢 Module 5: Pandas (Handling Missing Data & Data Cleaning)

📌 Part 1: Why Data Cleaning is Important?

👉 Real-world datasets are never perfect.

Datasets may contain:

- Missing values
- Wrong values
- Empty rows
- Duplicate data

👉 Before training AI models, data must be cleaned.

📌 Part 2: Theory Answers (IMPORTANT)

◆ 1. What is Missing Data?

👉 Missing data means some values are empty or unavailable in dataset.

👉 Example:

- Student marks missing
 - Salary not entered
 - Empty cells in Excel
-

◆ 2. Why Data Cleaning is Important?

👉 Dirty data causes:

- Wrong predictions
- Poor ML performance
- Incorrect analysis

- ◆ 3. What is isnull()?

👉 isnull() checks missing values in dataset.

- ◆ 4. What is dropna()?

👉 dropna() removes rows or columns containing missing values.

- ◆ 5. What is fillna()?

👉 fillna() replaces missing values with custom values.

📌 Part 3: Creating Dataset with Missing Values

```
import pandas as pd
import numpy as np

data = {
    "Name": ["Rahul", "Anu", "Arjun", None],
    "Marks": [85, np.nan, 78, 90],
    "City": ["Kochi", "Trivandrum", None, "Kollam"]
}

df = pd.DataFrame(data)

print(df)
```

📌 Part 4: Detect Missing Values

- ◆ Using isnull()

```
print(df.isnull())
```

- ◆ Count Missing Values

```
print(df.isnull().sum())
```

📌 Part 5: Removing Missing Data

- ◆ Drop Rows with Missing Values

```
print(df.dropna())
```

-
- ◆ Drop Columns with Missing Values

```
print(df.dropna(axis=1))
```

📌 Part 6: Filling Missing Values

- ◆ Replace with Custom Value

```
print(df.fillna("Unknown"))
```

-
- ◆ Fill Numeric Values

```
df["Marks"] = df["Marks"].fillna(df["Marks"].mean())
```

```
print(df)
```

👉 Missing marks replaced with average marks.

📌 Part 7: Duplicate Data Handling

- ◆ Check Duplicates

```
print(df.duplicated())
```

-
- ◆ Remove Duplicates

```
print(df.drop_duplicates())
```

📌 Part 8: Rename Columns

```
df.rename(columns={"Marks": "Score"}, inplace=True)
```

```
print(df)
```

📌 Part 9: Change Data Types

```
df["Score"] = df["Score"].astype(int)
```

```
print(df.dtypes)
```

📌 Part 10: Real-World Thinking

👉 In AI:

- Real datasets contain many missing values
- Cleaning improves model accuracy

👉 Example:

- Hospital records
 - Sales reports
 - Student attendance
-

📌 Part 11: Visualization of Missing Data

```
+-----+-----+-----+
| Name  | Marks | City   |
+-----+-----+-----+
| Rahul | 85    | Kochi  |
| Anu   | NaN   | Trivandrum |
| Arjun | 78    | NaN    |
| None  | 90    | Kollam |
+-----+-----+-----+
```

NaN / None → Missing Values

Tasks for Day 28

Task 1: Theory (Write Answers)

1. What is missing data?
 2. Why data cleaning is important?
 3. What is `isnull()`?
 4. What is `dropna()`?
 5. What is `fillna()`?
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Task 2: Missing Data Practice

1. Create dataset with missing values
 2. Detect missing values
 3. Count missing values
-

Task 3: Remove Missing Data

1. Drop rows with missing values
 2. Drop columns with missing values
-

Task 4: Fill Missing Data

1. Fill with text
 2. Fill with mean value
 3. Fill numeric columns
-

Task 5: Duplicate Handling

1. Create duplicate rows
2. Detect duplicates

3. Remove duplicates

Task 6: Column Operations

1. Rename columns
2. Change data type
3. Print data types